Highlights

Unsupervised Physiological State Clustering for Construction Worker Fatigue Monitoring: Comparing GMM and Hierarchical Clustering Across Temporal Resolutions

Carlos Alejandro Freitas Teixeira, Diogo Emanuel Pereira Martinho, Maria Goreti Marreiros, Rui Barros Garcia

- Unsupervised clustering identified fatigue and rest states.
- GMM and hierarchical clustering were compared at minute-level and window resolutions.
- Silhouette, BIC, and physiological interpretation guided the clusters.
- Original data better captured dynamic activity.
- External validation revealed differences between exercise, computer work, and sleep.